

Solutions — Practice Midterm (Alt B)

Problem 1

(a) $\frac{10!}{10^{10}} \approx 0.00036288$.

(b) $P(\geq 1 \text{ ten}) = 1 - (9/10)^{10} \approx 0.6513215599$.

Problem 2

$$P(I \mid BB) = \frac{0.25 \cdot \frac{1}{2}}{0.25 \cdot \frac{1}{2} + 0.75 \cdot \frac{1}{4}} = \frac{0.125}{0.3125} = 0.4.$$

Problem 3

$A \in \{0.22, 0.05, -0.08\}$, $B \in \{-0.12, 0.07, 0.00\}$, equally likely.

$$E[A] = 0.0633333, \quad E[B] = -0.0166667.$$

$$E[A^2] = 0.0191, \quad E[B^2] = 0.0064333 \Rightarrow \text{Var}(A) = 0.0150889, \quad \text{Var}(B) = 0.0061556.$$

$$E[AB] = -0.0076333 \Rightarrow \text{Cov}(A, B) = -0.0065778.$$

$$\text{Corr}(A, B) = \frac{-0.0065778}{\sqrt{0.0150889 \cdot 0.0061556}} \approx -0.6825.$$

For a 50/50 portfolio,

$$E[P] = 0.5(0.0633333 - 0.0166667) = 0.0233333, \quad \text{Var}(P) = 0.25(0.0150889 + 0.0061556 + 2(-0.0065778)) = 0.0020$$

so $\sigma_P \approx 0.04497$.

Problem 4

$X \sim \text{Bin}(63, 0.9)$. Then

$$P(X \leq 60) = 1 - \sum_{k=61}^{63} \binom{63}{k} (0.9)^k (0.1)^{63-k} \approx 0.9579.$$

Problem 5

For i.i.d. continuous variables $P(\text{record at } i) = 1/i$, so

$$E[Y_n] = \sum_{i=1}^n \frac{1}{i}.$$

For $n = 2$: $\text{Var}(Y_2) = \frac{1}{4}$. For $n = 3$: $E[I_2 I_3] = P(X_1 < X_2 < X_3) = 1/6$ so $\text{Cov}(I_2, I_3) = 0$, hence $\text{Var}(Y_3) = \frac{1}{4} + \frac{2}{9} = \frac{17}{36} \approx 0.4722$.

Problem 6

$$\Pi(x) = \begin{cases} -c(x), & 9 \leq x < 9.4, \\ 2.20 - c(x), & x \geq 9.4, \end{cases} \quad c(x) = \frac{x}{18} + 0.40.$$

With $f(x)$ triangular on $[9, 11]$ as given,

$$E[\Pi] = -\int_9^{9.4} c(x)(x-9) dx + \int_{9.4}^{10} (2.20 - c(x))(x-9) dx + \int_{10}^{11} (2.20 - c(x))(11-x) dx \approx 1.0684.$$